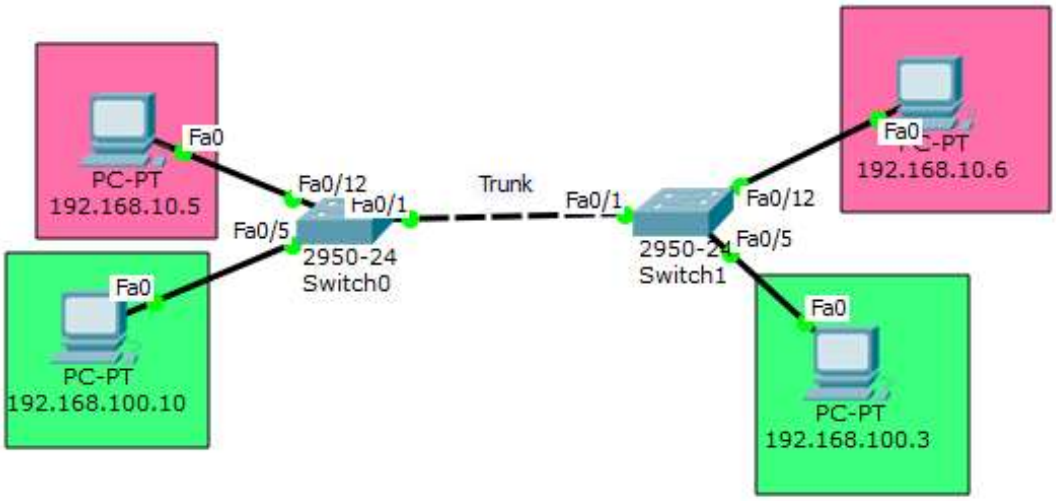
VTP Practical using Cisco Switches



Objective

- Configure VTP in a multi-switch environment using Cisco Catalyst 2950 switches.
- Create VLANs and verify their propagation via VTP.

VLAN Table:

VLAN Name	VLAN ID	Description	IP Address
HR	10	HR Department	192.168.100.0/24
Sales	20	Sales Department	192.168.10.0/24
IT	30	IT Department	192.168.20.0/24

Devices Required:

- 2× Cisco 2950 Switches (SW1, SW2, SW3)
- Ethernet cables

Connections:

- SW1 ↔ SW2 (FastEthernet 0/1 ↔ FastEthernet 0/1)

Procedure:

Step 1: Basic Configuration

On all switches:

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SWx   # Replace x with switch number
```

Step 2: Set VTP Domain and Mode

On SW1 (VTP Server):

```
SW1# configure terminal
SW1(config)# vtp domain BCOT
SW1(config)# vtp mode server
SW1(config)# vtp password bcot123
```

On SW2 (VTP Clients):

```
SW2# configure terminal
SW2(config)# vtp domain BCOT
SW2(config)# vtp mode client
SW2(config)# vtp password bcot123
```

Step 3: Configure Trunk Ports

On all trunking interfaces (SW1-SW2):

```
SWx# configure terminal
SWx(config)# interface fa0/1
SWx(config-if)# switchport mode trunk
SWx(config-if)# no shutdown
```

.Step 4: Create VLANs on Server

On **SW1 (VTP Server)**:

```
SW1# configure terminal
SW1(config)# vlan 10
SW1(config-vlan)# name HR
SW1(config)# vlan 20
SW1(config-vlan)# name IT
SW1(config)# vlan 30
SW1(config-vlan)# name ACCOUNTS
```

Step 5: Verify VTP Status and VLAN Propagation

On all switches:

```
SWx# show vtp status
SWx# show vlan brief
```

Expected Result:

- VLANs created on SW1 (HR, IT, ACCOUNTS) should appear automatically on SW2 .
- VTP configuration revision number should match on all switches.

Testing VTP Propagation

- Connect a PC to SW1 in VLAN 10.
- Connect another PC to SW2 in VLAN 10.
- Assign ports on SW1 and SW2 to VLAN 10:

```
interface fa0/5
switchport mode access
switchport access vlan 10
```

- Test connectivity between the two PCs (should be successful if VLANs propagated correctly).

Questions for Students

1. What is the purpose of the VTP revision number?
2. What happens if a client has a higher revision number than the server?
3. Can you create a VLAN directly on a VTP client?
4. What are the mode use of VTP modes?

Important Notes

- Ensure all switches are in the same VTP domain and have matching passwords.
- VTP version 1/2 is supported on 2950. Version 3 is not.
- Save configuration using `write memory` or `copy running-config startup-config`.